Hadamard-order FUNTFs for real ℓ_1^n

Agent lane 18 packet for run fa_banach_001

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Status

This is a candidate partial result for Question 8.2 of Chavez-Dominguez, Freeman, and Kornelson [1]. The source question asks whether real ℓ_1^n has a length N finite unit norm tight frame (FUNTF) for every $N \geq n$. The result below proves existence for all Hadamard-order lengths $N \geq n$, and in particular for every dyadic length $N = 2^m \geq n$. It does not settle arbitrary non-Hadamard lengths.

Source question evidence

The open problem appears on page 14, Section 8, Questions 8.1 and 8.2 of [1]. The relevant source crop is included below.

$\square$

8. Open problems

We showed in Section 4 that a large class of finite dimensional Banach spaces have finite unit norm tight frames of every length at least the dimension of the space. However, we do not have any examples where this is not possible.

Question 8.1: Does every n dimensional Banach space have a length N FUNTF for all $N \geq n$?

It seems very difficult to create a method of constructing FUNTFs of arbitrary size that works in any finite dimensional Banach space. Thus, it may be best to focus first on specific classical Banach spaces. We have shown in Section 4 that complex ℓ_1^n has a FUNTF of length N for all $n \leq N$ and in Section 7 that real ℓ_1^2, ℓ_1^3, and ℓ_1^4 each have FUNTFs of all lengths at least their dimension.

Question 8.2: Does real ℓ_1^n have a length N FUNTF for all $N \geq n$?

Note that for each n, there are only finitely many values of N for which we do not know the answer to the question above. Indeed, it follows from our results that for any $N \geq n(n-1)$ a FUNTF of length N does exist: if $N = n(n-1+m)+k$ with $0 \leq k \leq n-1$ and $m \geq 0$, we can take the union of k copies of a FUNTF of length $n+1$ and $n-1-k+m$ copies of a FUNTF of length n.

A finite dimensional Banach space X has a length N FUNTF if and only if a scalar multiple of the identity operator on X can be expressed as a sum of N normalized rank 1 projections. In the Hilbert space case, operators which can be written as sums of norm one rank one projections are characterized as positive operators with integer trace [KL04]. Operators on complex Banach

Question 8.2 asks: "Does real ℓ_1^n have a length N FUNTF for all $N \geq n$?" The same page notes that for each fixed n only finitely many values of N were then unknown, since the paper constructs lengths n and $n+1$ and then takes unions.

Proof Intuition

For real ℓ_1^n, a vector whose coordinates all have absolute value $1/n$ is normed by its coordinate sign vector in ℓ_∞^n. Thus each row of a sign matrix naturally gives a normalized rank-one projection. The only obstruction to tightness is cancellation of off-diagonal matrix entries in the sum of these projections. Orthogonality of the columns of the sign matrix is exactly the needed cancellation condition. A Hadamard matrix gives many such sign matrices at once, because any selection of n columns remains orthogonal.

Main partial result

Theorem 1 (Orthogonal sign columns give real ℓ_1 FUNTFs)**.** *Let $n \leq N$ be positive integers. Suppose that $H = (h_{r,i})_{1\leq r\leq N,\, 1\leq i\leq n}$ is an $N \times n$ matrix with entries in $\{-1,1\}$ and orthogonal columns:*

$$\sum_{r=1}^{N} h_{r,i}h_{r,k} = N\delta_{ik} \qquad (1 \leq i,k \leq n).$$

Then real ℓ_1^n has a length N FUNTF.

Proof. Let $(e_i)_{i=1}^n$ be the standard basis of ℓ_1^n, with biorthogonal functionals $(e_i^*)_{i=1}^n$ identified with the standard basis of ℓ_∞^n. For each row r of H, define

$$x_r = \frac{1}{n}\sum_{i=1}^{n} h_{r,i}e_i, \qquad x_r^* = \sum_{i=1}^{n} h_{r,i}e_i^*.$$

Then $\|x_r\|_1 = 1$, $\|x_r^*\|_\infty = 1$, and

$$x_r^*(x_r) = \frac{1}{n}\sum_{i=1}^{n} h_{r,i}^2 = 1.$$

Thus each $x_r^* \otimes x_r$ is a normalized rank-one projection in the sense used in the source paper.

Let $S : \ell_1^n \to \ell_1^n$ be the frame operator

$$S(x) = \sum_{r=1}^{N} x_r^*(x)x_r.$$

For a basis vector e_i,

$$S(e_i) = \sum_{r=1}^{N} h_{r,i}\left(\frac{1}{n}\sum_{k=1}^{n} h_{r,k}e_k\right) = \sum_{k=1}^{n}\left(\frac{1}{n}\sum_{r=1}^{N} h_{r,i}h_{r,k}\right)e_k = \frac{N}{n}e_i.$$

Hence $S = (N/n)I_{\ell_1^n}$, a scalar multiple of the identity. By the definition in [1], the sequence $(x_r, x_r^*)_{r=1}^N$ is a length N FUNTF for real ℓ_1^n. $\square$

Corollary 1 (Dyadic lengths)**.** *For every n and every m with $2^m \geq n$, real ℓ_1^n has a length 2^m FUNTF.*

Proof. The Sylvester construction gives a $2^m \times 2^m$ Hadamard matrix for every $m \geq 0$: start from $H_1 = (1)$ and set

$$H_{2M} = \begin{pmatrix} H_M & H_M \\ H_M & -H_M \end{pmatrix}.$$

Its columns are pairwise orthogonal. Select any n columns and apply Theorem 1. □

Corollary 2 (Hadamard orders)**.** *If a real Hadamard matrix of order N exists and $N \geq n$, then real ℓ_1^n has a length N FUNTF.*

Proof. Choose any n columns of the Hadamard matrix and apply Theorem 1. □

Relation to the source problem

The source paper proves that real ℓ_1^n has FUNTFs of lengths n and $n+1$, and that taking unions reduces Question 8.2 for fixed n to finitely many lengths. The dyadic corollary supplies, for every n, a uniform explicit length $2^{\lceil \log_2 n \rceil}$ with

$$n \leq 2^{\lceil \log_2 n \rceil} < 2n.$$

For values of n where this length is not already n or $n+1$, it fills one of the finite-gap lengths singled out by the source discussion. The theorem also covers every other known Hadamard order $N \geq n$.

This is still only a partial answer. It gives no construction for a length N unless an $N \times n$ sign matrix with orthogonal columns is available, and it does not decide the remaining non-Hadamard lengths.

Verifier report

The included script

`code/verify_hadamard_funtf.py`

constructs Sylvester matrices, chooses the first n columns, and checks in exact rational arithmetic that the frame operator produced above is $(N/n)I_n$ for $2 \leq n \leq 20$. From the packet directory, the command used was:

```
conda run --no-capture-output -n sandbox python
code/verify_hadamard_funtf.py
```

This computation is not a proof of the theorem; it is a reproducible sanity check for the construction and its normalization.

References

[1] J. A. Chavez-Dominguez, D. Freeman, and K. Kornelson, *Frame potential for finite-dimensional Banach spaces*, arXiv:1804.03677.